

Customer Churn Analysis

End-to-End Data & Machine Learning Portfolio Project

Telecom Industry · SQL Server · Power BI · Python Random Forest

7,043

Total Customers

1,869

Churned

26.5%

Churn Rate

411

New Joiners

This project delivers a complete end-to-end churn analysis pipeline for a telecom firm — from raw data ingestion in SQL Server, through Power BI dashboarding, to a Random Forest machine learning model that predicts future churners. The methodology is industry-agnostic and directly transferable to retail, finance, healthcare, and beyond.

Tech Stack

SQL Server

Power BI

Python

Scikit-learn

Pandas

Jupyter

1. Project Overview & Objectives

Customer churn — the rate at which customers stop doing business with a company — is one of the most critical KPIs in any subscription-based or service industry. This project demonstrates a production-ready churn analysis workflow built for a telecom dataset, covering every stage from data engineering to predictive modelling.

Project Goals

- Build a repeatable ETL pipeline in SQL Server with data quality checks and null handling.
- Create interactive Power BI dashboards segmented by Demographics, Geography, Account Info, and Services.
- Profile churner behaviour to surface actionable marketing campaign opportunities.
- Train a Random Forest classifier to predict which new customers are likely to churn.
- Visualise predicted churners in a dedicated Power BI report page.

Key Metrics Tracked

Metric	Definition	DAX Measure
Total Customers	All records in prod_Churn	COUNT(prod_Churn[Customer_ID])
Total Churn	Customers with status = Churned	SUM(prod_Churn[Churn Status])
Churn Rate	Churn / Total Customers	[Total Churn] / [Total Customers]
New Joiners	Status = Joined	CALCULATE(COUNT(...), Status="Joined")
Predicted Churners	ML model output = 1	COUNT(Predictions[Customer_ID]) + 0

Target Audience

While built on a telecom dataset, every technique demonstrated here — ETL design, dashboard architecture, and ML-driven churn prediction — applies equally to retail, SaaS, banking, healthcare, and any other industry where customer retention drives revenue.

2. Step 1 — ETL Process in SQL Server

2.1 Database Setup

A dedicated SQL Server database provides reliable, repeatable data loads with referential integrity — far more robust than a standalone Excel file for recurring ingestion.

```
CREATE DATABASE db_Churn
```

2.2 Staging Table Import

The raw CSV is imported into a staging table (stg_Churn) using the SQL Server Import Wizard (Task > Import > Flat File). Key settings: Customer_ID is set as the primary key; all other columns allow NULLs; BIT columns are cast to VARCHAR(50) to avoid import errors.

2.3 Data Exploration

Before cleaning, the distribution of key categorical fields and revenue by customer status is verified with exploratory queries:

```
SELECT Customer_Status, Count(*) as TotalCount, Sum(Total_Revenue) as TotalRev,
Sum(Total_Revenue)/(SELECT SUM(Total_Revenue) FROM stg_Churn)*100 AS RevPct FROM stg_Churn
GROUP BY Customer_Status
```

2.4 Null Handling & Production Load

A comprehensive NULL audit spans all 32 columns. Nullable service fields are defaulted to 'No' or 'None'; Churn_Category and Churn_Reason default to 'Others'. Clean records are inserted into the production table prod_Churn via SELECT ... INTO.

Column	Null Strategy
Value_Deal	ISNULL(Value_Deal, 'None')
Multiple_Lines	ISNULL(Multiple_Lines, 'No')
Internet_Type	ISNULL(Internet_Type, 'None')
Online_Security / Backup / etc.	ISNULL(col, 'No')
Churn_Category	ISNULL(Churn_Category, 'Others')
Churn_Reason	ISNULL(Churn_Reason, 'Others')

2.5 Power BI Views

Two SQL views are created to feed Power BI cleanly — one for historical churners and stayers, and one for new joiners awaiting prediction:

```
CREATE VIEW vw_ChurnData AS SELECT * FROM prod_Churn WHERE Customer_Status IN
('Churned','Stayed') CREATE VIEW vw_JoinData AS SELECT * FROM prod_Churn WHERE
Customer_Status = 'Joined'
```

3. Steps 2–4 — Power BI Transform & Visualisation

3.1 Power Query Transformations

After connecting to SQL Server, additional derived columns are created inside Power Query:

Churn Status (numeric)	if [Customer_Status] = "Churned" then 1 else 0 → cast to Integer
Monthly Charge Range	"< 20" / "20–50" / "50–100" / "> 100"
Age Group	"< 20" / "20–35" / "36–50" / "> 50" (+ sort index)
Tenure Group	"< 6 Months" through ">= 24 Months" (+ sort index)
prod_Services (unpivoted)	All service columns unpivoted → Attribute renamed Services, Value renamed Status

3.2 Dashboard Architecture

Summary Page

- Top KPI cards — Total Customers, New Joiners, Total Churn, Churn Rate %
- Demographics — Churn rate by Gender and Age Group
- Account Info — Churn rate by Payment Method, Contract type, Tenure Group
- Geographic — Top 5 States by Churn Rate
- Churn Distribution — Churn Category bar with Churn Reason tooltip
- Services — Internet Type churn rate + service usage heatmap

Churn Reason Tooltip Page

- Drill-through tooltip showing Churn Reason breakdown for any selected category

Churn Prediction Page

- Customer-level grid: ID, Monthly Charge, Total Revenue, Refunds, Referrals
- Demographics — Predicted churn count by Gender, Age Group, Marital Status
- Account Info — Predicted churn by Payment Method, Contract, Tenure
- Geographic — Predicted churn by State

4. Step 5 — Churn Prediction with Random Forest

4.1 Why Random Forest?

A Random Forest is an ensemble of decision trees, each trained on a random subset of the data and features. Predictions are determined by majority vote across all trees. This approach delivers strong accuracy, natural resistance to overfitting, and built-in feature importance rankings — ideal for a churn classification problem with mixed categorical and numerical inputs.

4.2 Pipeline Summary

#	Stage	Detail
1	Data Load	Import <code>vw_ChurnData</code> from SQL Server into Excel; read with <code>pandas.read_excel()</code> .
2	Pre-processing	Drop <code>Customer_ID</code> , <code>Churn_Category</code> , <code>Churn_Reason</code> . Encode 19 categorical columns with <code>LabelEncoder</code> . Map
3	Train / Test Split	80 / 20 split with <code>random_state=42</code> for reproducibility.
4	Model Training	<code>RandomForestClassifier(n_estimators=100, random_state=42).fit(X_train, y_train)</code> .
5	Evaluation	Confusion matrix + classification report (precision, recall, F1-score).
6	Feature Importance	Horizontal bar chart of top predictors (Tenure, Monthly Charge, Total Revenue, etc.).
7	New Predictions	Apply saved <code>LabelEncoders</code> to <code>vw_JoinData</code> , predict, filter to <code>Churned=1</code> , export CSV.

4.3 Key Code Snippets

Model training:

```
rf_model = RandomForestClassifier(n_estimators=100, random_state=42) rf_model.fit(X_train, y_train)
```

Evaluation output:

```
print(confusion_matrix(y_test, y_pred)) print(classification_report(y_test, y_pred))
```

Generating predictions on new joiners:

```
new_predictions = rf_model.predict(new_data) original_data['Customer_Status_Predicted'] = new_predictions original_data = original_data[original_data['Customer_Status_Predicted'] == 1] original_data.to_csv(r'Predictions.csv', index=False)
```

5. Insights & Business Recommendations

5.1 Churner Profile

Contract Type	Month-to-month customers churn at significantly higher rates than one- or two-year contract holders — the single strongest lever for retention.
Tenure	The first 6 months represent the highest churn risk window; onboarding programmes should target this cohort specifically.
Internet Type	Customers on Fiber Optic plans show elevated churn, suggesting price sensitivity or quality perception issues vs DSL alternatives.
Payment Method	Electronic check payers have the highest churn rate — incentivising auto-pay or credit card payments may reduce attrition.
Premium Support / Security	Customers without add-on services (Online Security, Device Protection, Premium Support) churn at higher rates.

5.2 Recommended Marketing Campaigns

Campaign	Target Segment & Action
Loyalty Upgrade Offer	Target month-to-month customers at the 3-month mark with a discounted annual contract offer.
Onboarding Nurture Sequence	Automate check-ins at weeks 2, 6, and 12 for all new joiners flagged by the ML model.
Bundle Promotion	Promote Online Security + Device Protection bundles to single-service customers via in-app and email.
Auto-Pay Incentive	Offer a monthly bill credit to customers switching from electronic check to auto-pay.
Win-Back Campaign	Re-engage recently churned customers with tailored offers based on their stated churn reason.

6. Technical Skills Demonstrated

Data Engineering

- Relational database design and management in Microsoft SQL Server
- ETL pipeline construction: staging → data quality → production table
- Null handling strategies with ISNULL / COALESCE
- SQL View creation for downstream BI consumption
- Flat-file CSV ingestion using the Import Wizard

Business Intelligence

- Power Query M transformations: column derivation, type casting, unpivoting
- DAX measure authoring (COUNT, SUM, CALCULATE, ratio measures)
- Multi-page dashboard design with drillthrough and tooltip pages
- KPI cards, bar/column charts, choropleth maps, matrix visuals
- User experience: page navigation, slicers, conditional formatting

Machine Learning

- Supervised classification with Scikit-learn Random Forest
- Label encoding of mixed categorical / numerical feature sets
- Train-test split and model evaluation (confusion matrix, F1-score)
- Feature importance extraction and visualisation with Seaborn
- Model persistence with joblib; prediction on unseen data

Python & Data Science

- Pandas for Excel ingestion, data manipulation, and CSV export
- NumPy for numerical operations and index sorting
- Matplotlib & Seaborn for exploratory and results visualisation
- Jupyter Notebook workflow for reproducible analysis
- Anaconda environment management and pip package installation

This project showcases an end-to-end production-grade analytics workflow — from raw data to predictive intelligence — entirely reproducible with open-source Python, SQL Server, and Power BI.